

When Scaling Forecasts Are Inadmissible: A Diagnostic for Error Decompositions

Author names and affiliations removed for double-anonymous peer review

Abstract

Organisations increasingly decide where to spend an artificial intelligence budget by asking which of a system’s failures the next model generation will remove on its own. The usual evidence is a decomposition: sort the errors into categories, fit a scaling curve to each, and read off which categories shrink fastest. We show this procedure is not merely noisy but inadmissible, and that the defect is structural rather than incidental. Because the categories partition the errors, their rates must sum to the total rate at every scale, and independently fitted curves cannot satisfy that identity once their exponents differ, which is the only reason to fit them separately. The largest category exponent necessarily exceeds the total’s, so the parts outgrow the whole and the forecast describes an impossibility beyond a computable scale. On our own measurements the parts reach 1.27 times the whole at one billion parameters and 2.82 times at one trillion. The threshold sits at 174M parameters, below the range our earlier analysis had forecast into. An independent model family reproduces the defect at 243M. We contribute a diagnostic that needs only the exponents a study already publishes, a closed remedy that fits the composition on the simplex and is 5.4% more accurate in range, and the finding that an inadmissible fit voids negative conclusions as well as forecasts, which we demonstrate by retracting one of our own. For managers the practical yield is a cheap, auditable test of whether a supplier’s roadmap claim is arithmetic or assertion.

Keywords: scaling laws; error decomposition; design science; model evaluation; artificial intelligence governance; forecast admissibility

1 Introduction

A firm running a document-processing system, a coding assistant, or a clinical triage aid faces a recurring question with a large budget attached. The system fails in several distinguishable ways. Some of those failures the next model generation will probably remove without anyone doing anything. Others it will not. Money spent mitigating the first kind is wasted, and money withheld from the second kind buys a defect that persists through every upgrade. Deciding which is which is a forecasting problem, and the field has a standard way of answering it: partition the observed errors into categories, fit a scaling curve to each category, and compare the exponents.

The procedure is appealing because it is cheap and because it appears to convert an argument about the future into a measurement. It is also, as we show here, inadmissible whenever the exponents differ, which is the only circumstance in which anybody bothers to compute them. The categories partition the errors, so their rates must sum to the total error rate at every scale. Five curves fitted one at a time do not know this and cannot enforce it. The category with the shallowest decline eventually dominates a total that is falling faster than it is, the parts overtake the whole,

and the forecast is then assigning more failures to one category than occur altogether. This is not a wide confidence interval. It is an arithmetic impossibility, and it means the extrapolation stopped reporting the exponents’ implications somewhere below the scale at which it was read.

We came to this the embarrassing way. A companion study of ours (Anonymous, 2026) decomposed the illegal moves made by chess transformers, fitted a curve per category, and projected to one billion parameters. That projection was inadmissible, and the threshold at which it became so, 174M parameters, lay well below the scale we had read it at. The fitted curves had passed every quality gate we had, agreed closely with the data in range, and were still describing something that cannot happen. We report the diagnosis, the remedy, and our own retraction together, because the sequence is the argument: this failure is invisible to the checks practitioners actually run.

Why this matters beyond the laboratory. The decomposition-and-extrapolate pattern has escaped the research literature. It appears in vendor roadmaps (“this class of error is on a steep curve”), in internal capability planning, in procurement comparisons, and in assurance arguments submitted to auditors. A manager reading such a claim currently has no way to check it other than trusting the analyst, because the defect is not visible in any of the artefacts normally shown: the fit looks excellent, the residuals are small, the intervals are narrow. The diagnostic we provide changes that. It needs only the exponents a study already prints, it runs in milliseconds, and it returns a scale past which the forecast is not entitled to its own numbers. That converts a question of analytic authority into a question anyone can audit.

Research questions and objectives.

- RQ1** Is the failure of independently fitted category curves to respect the partition identity incidental to particular data, or structural?
- RO1** Establish the condition under which the parts must outgrow the whole and derive the scale at which a given decomposition becomes inadmissible (Section 3).
- RQ2** Can the defect be detected from what published studies already report, without access to their data?
- RO2** Design and implement a diagnostic that consumes only reported exponents and intercepts (Sections 4 and 5).
- RQ3** Does a closed reformulation restore admissibility without costing accuracy?
- RO3** Fit the composition on the simplex and compare in-range accuracy against the uncoupled baseline (Section 7).
- RQ4** What does an inadmissible fit invalidate, beyond the forecast itself?
- RO4** Test whether conclusions drawn against such a fit survive, using a hypothesis we had previously rejected on exactly that basis (Section 7).

Contributions. The contribution is a diagnostic artifact together with the design knowledge it carries, in three pillars. First, an admissibility condition for per-category scaling forecasts, with a computable breakdown scale, which shows the defect is forced rather than accidental. Second, a closed parameterisation that removes it and is more accurate in range than the practice it replaces, so adoption costs nothing. Third, evidence that inadmissible fits void negative results as well as positive ones, demonstrated by retracting a rejection of our own that turned out to rest on nothing.

2 Related Work

Scaling laws and what is fitted to them. Power-law relationships between model scale and loss (Kaplan et al., 2020) and their compute-optimal refinement (Hoffmann et al., 2022) established the practice of extrapolating fitted curves to scales not yet trained. Those studies fit a single aggregate quantity, for which no partition identity exists and the criticism here does not apply. The difficulty arrives when the same machinery is applied category by category to a decomposed error, which is what practitioners want because it is category-level behaviour that determines where to spend. We differ from this literature by asking not how accurate such fits are but whether they are admissible at all.

Metric artefacts in scale measurement. Schaeffer et al. (2023) showed that apparently discontinuous capability gains can be manufactured by the choice of metric, and that linear or continuous metrics produce smooth curves from the same model outputs. That result and ours are the same genre of criticism, that a conclusion about models turned out to be a conclusion about the measurement apparatus, and they are complementary in mechanism: theirs concerns the shape imposed by a discontinuous scoring function, ours the constraint violated by fitting the parts of a whole independently. A study can satisfy their recommendation completely, using a perfectly continuous metric, and still produce the inadmissible forecast we describe.

Design science and the evaluated artifact. We frame the contribution as design-science research in the sense of Hevner et al. (2004), following the process of Peffers et al. (2007) and positioning the output as an improvement-type contribution in the terms of Gregor and Hevner (2013): a known problem, forecasting where scale will help, addressed by a new solution whose transferable content is the admissibility condition rather than the code that computes it. The artifact is evaluated against measurements from two independent model families, one of which we did not produce.

The instrument domain. Our measurements come from chess transformers, where a rules engine supplies an exact and cheap oracle for whether an output is legal and for which constraint it violated (Karvonen, 2024). That is why the domain is used here: it yields an error decomposition that is unambiguous and not itself a modelling choice. Nothing in the diagnostic is specific to it, and Section 9 sets out where else the pattern applies.

3 Design Problem and Requirements

The problem, stated plainly. A study reports, for each of k error categories, a fitted relationship $r_i(N) = A_i N^{a_i}$ between model scale N and the category’s rate, and a separate fit $T(N) = BN^b$ for the total error rate. A reader wants to know what the composition looks like at some N beyond the fitted range. The categories partition the errors, so the identity

$$\sum_{i=1}^k r_i(N) = T(N) \quad \text{for all } N \tag{1}$$

holds of the quantities being estimated. It does not hold of the estimates.

Table 1: Design principles and their realisation in the diagnostic.

Design principle	Realisation
DP1. Consume only what is published	The check takes exponents and intercepts, the figures a study already prints, so it can be applied to work whose data are unavailable.
DP2. Separate the verdict from the remedy	The diagnostic reports admissibility and a breakdown scale without requiring the analyst to adopt any particular replacement model.
DP3. Report a scale, not a verdict alone	A decomposition is not simply good or bad; it is usable up to a threshold, so the output is the threshold.
DP4. Make the admissible alternative cost nothing	The closed reformulation is chosen so that in-range accuracy does not decline, removing the incentive to skip it.
DP5. Fail loudly on inputs that cannot bear a fit	A category with a zero rate at any scale has no exponent; the implementation refuses rather than flooring the value and inventing a slope.
DP6. State what is invalidated	Output distinguishes what an inadmissible fit costs, which includes conclusions drawn against it, not only its forecasts.

The admissibility condition. Let $a^* = \max_i a_i$. The sum of the parts is asymptotically proportional to N^{a^*} , the whole to N^b , so the ratio of parts to whole grows as N^{a^*-b} and diverges precisely when $a^* > b$. That inequality is not a coincidence to be checked case by case. Fitted within the observed range, b tracks the rate-weighted mean of the a_i , because the total is the rate-weighted sum of the parts there. A maximum is at least a weighted mean, with equality only if every a_i is identical. Hence:

Any per-category scaling decomposition whose fitted exponents are not all equal must violate Equation 1 beyond a finite scale, and that scale is computable from the fitted coefficients alone.

Since exponents that are all equal would make the decomposition pointless, every decomposition worth reporting is inadmissible somewhere. The question is only whether the scale at which it becomes so lies above or below where the forecast was read. Our own values illustrate the mechanism: a^* is -0.112 , the rate-weighted mean of the part exponents is -0.266 , and the separately fitted b is -0.267 , giving a divergence rate of $+0.154$.

Design principles. Table 1 states the principles the artifact realises. They are stated so as to transfer: none mentions chess, error legality, or any property of our instrument.

In scope and out of scope. In scope: decompositions in which categories partition a measured total, each fitted as a power law in a common scale variable. Out of scope: single aggregate scaling laws, for which no partition identity exists and the criticism does not apply; categories that overlap rather than partition, for which a different bound is needed; and the question of whether any particular fitted exponent is correct, which this work does not address and does not need to. An inadmissible forecast can rest on exponents that are individually excellent, which is precisely what makes the defect hard to see.

4 Artifact Architecture

The artifact has three separable parts, deliberately, so that an analyst may adopt one without the others.

The checker. A module that takes a dictionary of category coefficients and a total, and returns the divergence rate $a^* - b$, an admissibility verdict, the breakdown scale at a chosen tolerance, and the overspend ratio at any scales of interest. It depends on one numerical library and nothing else, including nothing from the study that produced our measurements, so that it can be pointed at published work in unrelated fields.

The estimator. A convenience path that fits the coefficients from raw measurements rather than requiring them to be transcribed, applying DP5: a category whose rate is zero at some scale is refused, with an explanation, rather than silently floored.

The closed model. The remedy, which parameterises the composition as a softmax linear in $\log N$ and multiplies it by a separately fitted total. Shares sum to one at every scale for every parameter value, so Equation 1 holds by construction and no setting of the parameters can produce an impossible forecast. The total keeps an ordinary power law, which is legitimate because a total is not a part of anything.

Tolerance. Admissibility is asymptotic, but in-range fit noise makes a strict threshold useless, so the breakdown scale is defined at a tolerance: the scale at which the parts exceed the whole by more than a stated fraction, ten percent by default. The search starts at the top of the observed range, because the ratio of parts to whole is U-shaped, large at small scales where the steep categories dominate and large at large scales where the shallow one does. Searching upward from zero finds the wrong arm and reports a threshold beneath the data, which is a mistake we made and fixed.

5 Method and Measurements

Instrument. Measurements come from 18 decoder-only transformers spanning 3.4M to 39.9M parameters, 6 sizes at 3 seeds each, trained on human chess games and evaluated without search. Every model output from ply twenty onward is checked by a rules engine and, when illegal, assigned to exactly one category by what made it illegal: no piece on the source square, an opponent’s piece there, the mover’s own piece on the destination, a geometric violation, or a move that is legal in every respect except that it leaves the king in check. The classes are ordered by how much of the board must be consulted to detect them, which is why their exponents differ and why the decomposition is informative in the first place.

Independent replication. The second family is a publicly released ladder of character-level chess models trained by another laboratory on a different corpus with different code (Karvonen, 2024), spanning a 38-fold parameter range. Its notation does not name a source square, so its taxonomy is coarser than ours; we do not attempt to match the taxonomies when testing admissibility, because the diagnostic does not require it. Each family is checked against its own partition.

Provenance. Every adapter between an external model and our measurements fails silently rather than with an error, so each was verified and the checks recorded: a reconstructed 32-character vocabulary, move-boundary alignment verified three ways at 4,081 sites with 0 disagreements, and a forward pass choosing a legal move at 672 of 672 positions in a third model. No number in this manuscript is typed by hand; each enters through a generated macro tagged with the results file it came from.

Computing environment. All runs fit on a single NVIDIA GeForce RTX 3050 Laptop GPU with 4 GiB of device memory, in a host with 15 GB of physical memory visible to the operating system, under Windows 11, using Python 3.12.10 with PyTorch 2.5.1+cu121, NumPy 2.4.6 and SciPy 1.18.1. The memory ceiling, rather than a judgement about what was worth measuring, is what sets the top of our ladder, which is why the independently trained family carries weight in Section 7.

6 Illustrative Walkthrough

An analyst has fitted five category curves and a total to the ladder above, and wishes to state what the composition will look like at one billion parameters. The fits are good: each category declines monotonically across the ladder, and the category exponents are separated well beyond their intervals, from -0.574 for the most local category to -0.112 for the least, a paired differential of $+0.461$ $[+0.259, +0.599]$. Nothing in the residuals suggests trouble.

Running the check takes the ten fitted coefficients and returns: a^* is -0.112 against a total exponent of -0.267 , a divergence rate of $+0.154$, and a breakdown scale of 174M parameters at a ten percent tolerance. The intended forecast scale is one billion. The forecast is therefore being read roughly six times past the point at which the decomposition stops being arithmetic, and the overspend there is 1.27 $[1.13, 1.57]$, above one in 100% of resamples of the ladder.

The analyst now has two usable options and one unusable one. The unusable option is to widen the intervals, which does not help, because the problem is not that the estimate is imprecise. The first usable option is to report the forecast only up to 174M parameters and say so. The second is to refit the composition on the simplex, which restores admissibility at every scale and, on these data, improves in-range accuracy. Section 7 reports what that refit gives.

7 Evaluation

Does the defect occur in practice, and how badly? Table 2 gives the ratio of independently extrapolated parts to the independently extrapolated whole. At the top of our observed range the ratio is 1.02, which is fit noise and unobjectionable. One order of magnitude beyond it the parts have overtaken the whole, and three orders beyond it they are nearly three times the total they are part of.

Does it replicate on a family we did not train? Yes, and at a similar magnitude. Refitting the independent ladder over its own 4 usable categories gives a^* -1.044 against a total of -1.278 , a divergence rate of $+0.234$, an overspend of 1.28 at one billion parameters, and a breakdown scale of 243M. A different tokenisation, a different taxonomy, a different corpus and a different laboratory reproduce the defect within the same order of magnitude, which is what Section 3 predicts and what makes this a property of the method rather than of our data. One of that family’s categories

Table 2: Parts-to-whole ratio under independently fitted category curves. A value above one is not a wide estimate but an impossibility. Interval from resampling the ladder’s size rungs.

Scale	Parts / whole	Interval
Top of observed range (39.9M)	1.02	n/a
One billion parameters	1.27	[1.13, 1.57]
One trillion parameters	2.82	n/a

is excluded because its rate is zero at two of three checkpoints; an exponent fitted there would be an artefact of the floor, and DP5 makes the implementation refuse it rather than produce one.

Does the remedy cost accuracy? No, it improves it. Fitted in range across the 18 runs, the closed model attains a root-mean-square error of 0.01251 in share space against 0.01322 for the uncoupled fits, an improvement of 5.4%. This matters for adoption rather than for truth: an admissible model that fitted worse would face a trade-off argument, and this one does not. Under the closed model the least local category’s share rises from 0.444 at the top of our range to 0.588 at one billion and 0.769 at one trillion, approaching but never exceeding 0.871, while the most local category falls to 0.001. The substantive ordering the decomposition was meant to establish survives. What changes is that the numbers are now ones the categories can jointly satisfy.

The same discipline outside the partition. A quantity confined to the unit interval needs a link that respects the confinement even when it is not a member of a partition. The share of errors occurring despite a correctly decoded local state, which cuts across the categories rather than belonging to them, was observed between 0.148 and 0.321. Fitted on the logit it reaches 0.596 at one billion and 0.951 at one trillion, remaining a proportion throughout. Fitted linearly in the log, as the categories originally were, it would not.

What an inadmissible fit invalidates. This is the evaluation we did not expect to run. While preparing this manuscript we tested a hypothesis that failure composition is governed by a model’s achieved competence rather than by its size, rejected it on the grounds that the fitted shares exceeded one, and recorded the rejection. That rejection was worthless. Predicting an impossibility is what the uncoupled form does everywhere, so its disagreement with the data carried no evidence about the hypothesis under test. We repeated the comparison admissibly, fitting the composition on the simplex in competence coordinates, matching the two families down to the three categories both taxonomies share, and predicting the independent family’s checkpoints entirely out of sample. The hypothesis is genuinely refuted: mean total variation distance 0.490, against 0.100 for size coordinates and 0.115 for the baseline of predicting our own mean composition regardless of input. The conclusion survived. The original argument for it did not. An inadmissible fit cannot be used to reject a hypothesis, because a model that predicts impossibilities everywhere cannot be embarrassed by data anywhere. We have not seen this stated, and we had it wrong ourselves until we checked.

Honest negative results. Two further ideas were tested and are reported because they failed. We predicted from the closed model that error diversity should collapse with scale, since a softmax converges to its steepest component, and measured the normalised entropy of the composition. It declines on our ladder, with a slope whose bootstrap interval excludes zero, and it rises on the independent ladder. One ladder is not a result, and we do not report entropy as a scaling signature.

We also computed the scale at which the least local category overtakes the most local one, intending it as a headline, and found it lies below our smallest model, so it describes the past rather than the future. Neither appears in our conclusions.

8 Discussion

What the integration buys. Each part of the artifact is individually unremarkable. A ratio of two fitted curves is arithmetic; a softmax is standard; refusing to fit a zero is hygiene. What the combination buys is that a practitioner can act on the finding without first believing it. The checker tells an analyst that a forecast is inadmissible using only what the analyst already has, and the closed model gives somewhere to go that costs nothing in accuracy. A criticism that arrives with a remedy priced at zero is adopted; one that arrives as a warning is filed.

A design-choice defence. The softmax is one admissible parameterisation among many, and we do not claim it is the right one. Any link that maps the real line onto the simplex would enforce Equation 1, and different links would give different numbers beyond the observed range while agreeing that the uncoupled numbers are inadmissible. We chose the softmax because it is the minimal change from a log-linear fit, reducing to one when there are two categories, so an analyst can adopt it without adopting a new modelling philosophy. The transferable knowledge is the admissibility condition, not the link.

Why the defect survives ordinary quality control. Our uncoupled fits passed every gate we had. The residuals were small, the exponents were well separated, the intervals were narrow, and the category rates declined monotonically. None of those checks is sensitive to the violated identity, because each examines one category at a time and the identity is a statement about the categories jointly. This is the practical lesson for anyone maintaining an evaluation pipeline: goodness-of-fit is computed within a series, and the constraint lives across them.

Design-knowledge contribution. Stated at the level at which it transfers: when a quantity is decomposed and the components are modelled separately, the identity that made the decomposition meaningful becomes an unenforced constraint, and extrapolation is where the absence of enforcement becomes visible. The remedy is to move the constraint into the parameterisation rather than to check it after the fact, and the diagnostic value of checking it after the fact is that it tells you how far the unconstrained model could be trusted. This applies wherever errors are decomposed by capability, benchmark subset, severity, customer segment, or failure type, and the test costs one function call.

9 Implications for Individuals, Organisations, and Society

For the people affected by the systems these forecasts govern. The forecasts in question decide which defects get fixed and which are left to be cured by the next model. That decision is felt by whoever is on the receiving end of the defect that was left: the applicant whose document is misread, the patient whose triage note is misfiled, the small trader whose claim is misclassified. When a category of failure is forecast to disappear with scale, mitigation for it is deferred, and the deferral lasts as long as the forecast is believed. An inadmissible forecast systematically favours deferral, because the categories it inflates are the shallow ones, which are exactly the ones described as persistent. The people who carry the cost of an over-optimistic roadmap are rarely the people

who wrote it, and they have no way to contest an analysis they cannot see. A test that runs on published coefficients gives an advocate, a regulator, or an ombudsman something concrete to ask for.

For managers and organisations. The practical reframing is from a number to be received into a claim to be governed. A supplier asserting that a failure class is on a steep curve is making a forecast with an admissibility property, and that property can be checked before the contract is signed rather than after the upgrade disappoints. We would put the question directly into evaluation practice: ask for the fitted coefficients, run the check, and require that the forecast scale lie below the breakdown scale. This is cheap enough to be routine, it needs no access to the supplier’s data or weights, and it is auditable, so the reasoning survives the departure of whoever did it. It also disciplines internal planning, where the same pattern is used to decide whether to staff a mitigation team or wait for the next generation. Waiting is a decision with a cost, and it should not rest on arithmetic nobody checked.

Strategic implications. Treat forecast admissibility as an ownership question. An organisation that can audit capability claims in-house is not dependent on its suppliers’ analysts for the most consequential input into its own roadmap, and it is not exposed when a vendor relationship ends. The artifact is small, has one dependency, and requires no proprietary data, so this is genuinely available to organisations that could not build an evaluation science function. We also read it as an accountability asset rather than an overhead. A firm that can show it tested the admissibility of the forecasts underlying a deployment decision is in a materially stronger position with auditors and regulators than one that can show only that it believed a vendor, and the record is created as a by-product of doing the analysis properly.

For society and digital equity. The ability to check a scaling claim should not be a function of having a research team. At present, evaluating whether a capability forecast is sound requires either the underlying data, which is usually unavailable, or the standing to be believed when disagreeing, which is distributed very unevenly. A diagnostic that consumes only published coefficients moves this from the second category to the first: it is checkable by a civil-society organisation, a journalist, a procurement officer in a small public body, or a graduate student, on ordinary hardware. We note that all measurements reported here were produced on a single consumer laptop graphics card, which is not incidental to that claim. We also record what the artifact deliberately does not do. It does not rank models, score vendors, or certify anything. It answers one narrow question, whether a forecast is entitled to the numbers it reports, and it leaves the judgement of what to do about that with the people accountable for the decision.

10 Limitations

- L1. One instrument domain.** Both families are chess transformers, chosen because a rules engine gives an exact and unambiguous error partition. The admissibility argument is arithmetic and does not depend on the domain, but the empirical demonstration does.
- L2. The closed model is coherent, not validated.** The softmax guarantees the forecast is arithmetically possible, which the uncoupled fits are not. It does not make it observed. Every scale beyond 39.9M is extrapolation, and a different admissible link would give different numbers.

- L3. Three checkpoints in the independent family.** The replication supports directional agreement on admissibility rather than a precise second estimate of any exponent, and its coefficients carry correspondingly little precision.
- L4. Depth and width are confounded.** Both ladders grow depth and width together, so an exponent in parameters isolates neither.
- L5. Partitions only.** The condition is derived for categories that partition a total. Overlapping categories are common in practice and need a different bound, which we have not derived.
- L6. No audit of the published literature.** We show the defect is forced and demonstrate it on two families. We have not surveyed published decompositions to establish how often forecasts are read beyond their breakdown scale, which is the question a practitioner would most like answered.
- L7. The tolerance is a convention.** Ten percent is a defensible default and not a derived quantity; a different tolerance moves the reported breakdown scale though not the verdict.

11 Avenues of Future Research

- FR1** Apply the diagnostic in a domain whose error partition comes from a non-game oracle, such as program execution traces or spreadsheet recalculation, where a checker can say not only that a step is wrong but which constraint it violated, to establish that the empirical demonstration is not an artefact of chess.
- FR2** Compare admissible links, including the additive-logratio and multinomial-logit families, on their out-of-range behaviour, so that an analyst choosing among them has evidence rather than only the guarantee they share.
- FR3** Extend the independent replication to a family with more checkpoints, so that admissibility can be tested against exponents estimated with usable precision rather than only in direction.
- FR4** Separate depth from width along a ladder that varies them independently, and determine whether the divergence rate is a property of parameter count or of one of its components.
- FR5** Derive the corresponding bound for overlapping categories, which would extend the check to severity scales and multi-label failure taxonomies as used in industrial evaluation.
- FR6** Conduct a survey of published per-category scaling forecasts, applying the diagnostic to their reported coefficients, to estimate how frequently forecasts are read beyond their own breakdown scale.
- FR7** Study the tolerance itself, relating it to the sampling error of the underlying rate estimates so that the breakdown scale can be reported with a principled rather than a conventional threshold.

12 Conclusion

Decomposing a system’s errors and fitting a scaling curve to each category is a reasonable thing to want to do, and the resulting forecast is used to decide where money and attention go. We have shown that the forecast is inadmissible whenever the fitted exponents differ, which is whenever the exercise is worth performing, and that the scale at which it becomes so can be computed from the coefficients a study already publishes. On our own measurements that scale was 174M parameters and we had read a forecast at one billion; an independently trained family breaks down at 243M. The remedy costs nothing, improving in-range accuracy by 5.4%, and the diagnostic costs a function call.

The finding we would most want carried forward is the one we obtained by being wrong. An inadmissible fit does not merely produce forecasts that are too confident; it produces a model that

cannot be contradicted by evidence, and so it silently voids the negative conclusions drawn against it as well as the positive ones drawn from it. We found this because a hypothesis we had rejected turned out to have been rejected by nothing. The people this ultimately matters for are not the analysts. They are the users of systems whose known defects were left unrepaired because a forecast said scale would take care of them, and who had no way to ask whether the arithmetic held.

Declarations

Disclosure statement. Removed for double-anonymous peer review.

Funding. Removed for double-anonymous peer review.

Data availability statement. The code, the measurement artefacts, and the generated manuscript sources will be made available upon acceptance.

AI usage declaration. During the preparation of this manuscript the authors used a generative AI tool to assist with language editing and refinement of the text. The authors reviewed and edited all output and take full responsibility for the content of the paper. Generative AI was not used to generate or analyse the study’s data, which were produced entirely by the evaluation pipeline described in the paper.

References

- Gregor, S. and Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly*, 37(2):337–355.
- Hevner, A. R., March, S. T., Park, J., and Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1):75–105.
- Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Driessche, G. v. d., et al. (2022). Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*.
- Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., and Amodei, D. (2020). Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*.
- Karvonen, A. (2024). Emergent world models and latent variable estimation in chess-playing language models. In *First Conference on Language Modeling (COLM)*.
- Peppers, K., Tuunanen, T., Rothenberger, M. A., and Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*, 24(3):45–77.
- Schaeffer, R., Miranda, B., and Koyejo, S. (2023). Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*.